



Worksheet 1

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Subject Name: Internet of Things

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1. Aim/Overview of the practical:

Interface an Arduino Uno with LEDs to design the different patterns and follow the different cases. Use 8 different color LEDs and name them T1 to T8 and use resistors of suitable values.

- T1, T2, T3, T4 are in ON condition for 3 secs and T5, T6, T7 and T8 are in OFF condition for 2 secs.
- T1, T2, T3 and T4 are in OFF condition for 2.5 secs and T5, T6, T7 and T8 are in ON conditions for 3.5 secs.
- T1, T3, T5 and T7 are in ON condition for 4 secs and T2, T4, T6 and T8 are in OFF conditions for 2 secs.
- T1, T3, T5 and T7 are in OFF conditions and T2, T4, T6 and T8 are in ON conditions simultaneously.

2. Apparatus (For applied/experimental sciences/materials based labs):

Hardware Requirements

- Arduino Uno
- 8 LEDs (different colors)
- $8 \times 220 \Omega$ (or 330Ω) resistors
- Breadboard
- Jumper wires
- USB cable

Software requirements

- Arduino IDE
- Tinkercad Circuits

3. Circuit Diagram (TinkerCad):

- T1, T2, T3, T4 are in ON condition for 3 secs and T5, T6, T7 and T8 are in OFF condition for 2 secs.

```
int T1 = 2;
```

```
int T2 = 3;
```



```
int T3 = 4;

int T4 = 5;

int T5 = 6;

int T6 = 7;

int T7 = 8;

int T8 = 9;

void setup() {

    pinMode(T1, OUTPUT);

    pinMode(T2, OUTPUT);

    pinMode(T3, OUTPUT);

    pinMode(T4, OUTPUT);

    pinMode(T5, OUTPUT);

    pinMode(T6, OUTPUT);

    pinMode(T7, OUTPUT);

    pinMode(T8, OUTPUT);

}

void loop() {

    digitalWrite(T1, HIGH);

    digitalWrite(T2, HIGH);

    digitalWrite(T3, HIGH);

    digitalWrite(T4, HIGH);

    digitalWrite(T5, LOW);
```

```
digitalWrite(T6, LOW);

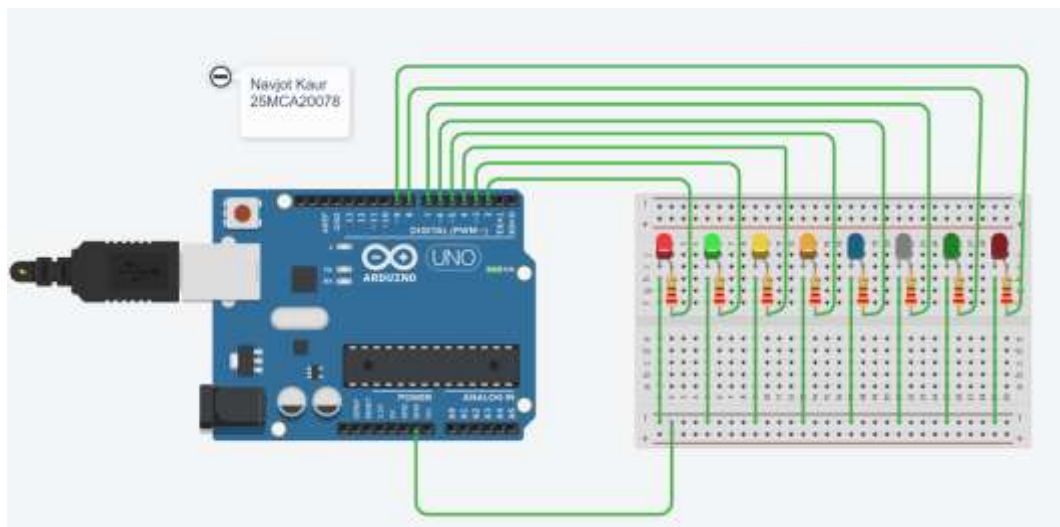
digitalWrite(T7, LOW);

digitalWrite(T8, LOW);

delay(3000);

delay(2000);

}
```



b) T1, T2, T3 and T4 are in OFF condition for 2.5 secs and T5, T6,T7 and T8 are in ON conditions for 3.5 secs.

```
int T1 = 2;
int T2 = 3;
int T3 = 4;
int T4 = 5;
int T5 = 6;
int T6 = 7;
int T7 = 8;
int T8 = 9;

void setup() {
  pinMode(T1, OUTPUT);
  pinMode(T2, OUTPUT);
  pinMode(T3, OUTPUT);
  pinMode(T4, OUTPUT);
  pinMode(T5, OUTPUT);
  pinMode(T6, OUTPUT);
```

```
pinMode(T7, OUTPUT);
pinMode(T8, OUTPUT);
}
```

```
void loop() {
```

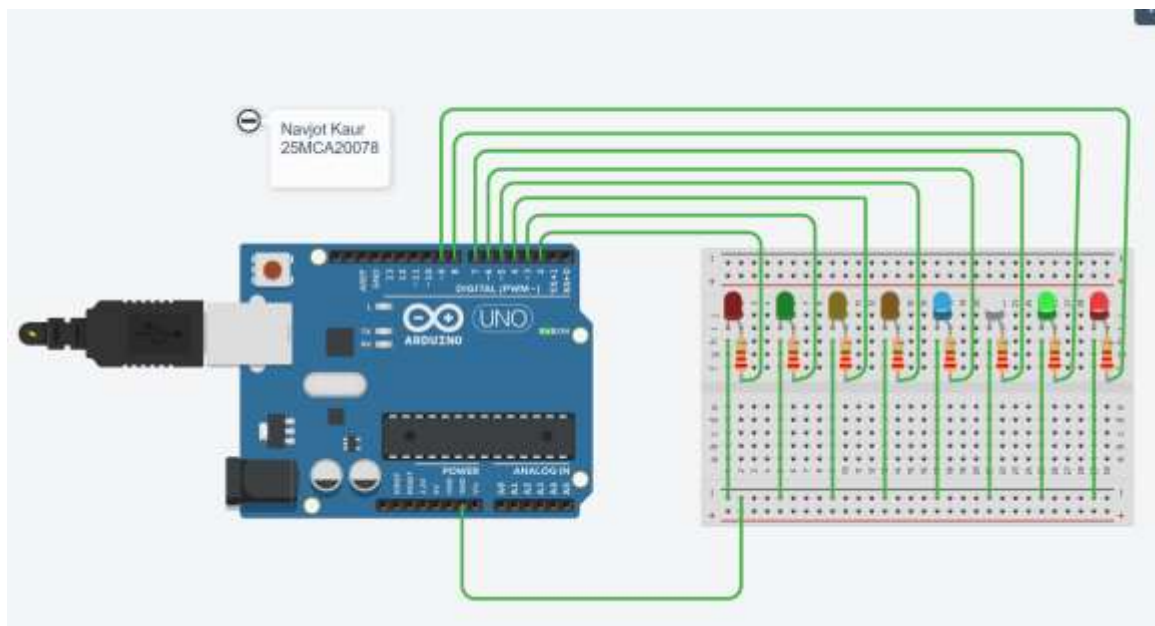
```
digitalWrite(T1, LOW);
digitalWrite(T2, LOW);
digitalWrite(T3, LOW);
digitalWrite(T4, LOW);
```

```
digitalWrite(T5, HIGH);
digitalWrite(T6, HIGH);
digitalWrite(T7, HIGH);
digitalWrite(T8, HIGH);
```

```
delay(2500);
```

```
delay(3500);
```

```
}
```



c) T1, T3, T5 and T7 are in ON condition for 4 secs and T2, T4, T6 and T8 are in OFF conditions for 2 secs.

```
int T1 = 2;
```

```
int T2 = 3;
```



```
int T3 = 4;
```

```
int T4 = 5;
```

```
int T5 = 6;
```

```
int T6 = 7;
```

```
int T7 = 8;
```

```
int T8 = 9;
```

```
void setup() {
```

```
    pinMode(T1, OUTPUT);
```

```
    pinMode(T2, OUTPUT);
```

```
    pinMode(T3, OUTPUT);
```

```
    pinMode(T4, OUTPUT);
```

```
    pinMode(T5, OUTPUT);
```

```
    pinMode(T6, OUTPUT);
```

```
    pinMode(T7, OUTPUT);
```

```
    pinMode(T8, OUTPUT);
```

```
}
```

```
void loop() {
```

```
    digitalWrite(T1, HIGH);
```

```
    digitalWrite(T2, LOW);
```

```
    digitalWrite(T3, HIGH);
```

```
    digitalWrite(T4, LOW);
```

```
    digitalWrite(T5, HIGH);
```

```
digitalWrite(T6, LOW);

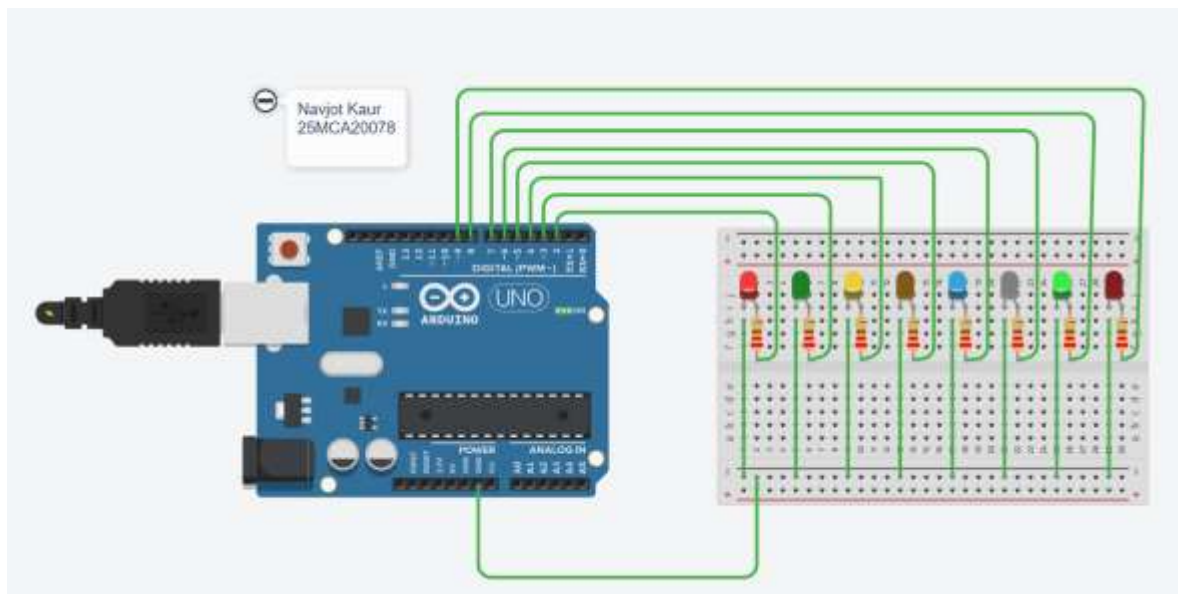
digitalWrite(T7, HIGH);

digitalWrite(T8, LOW);

delay(4000);

delay(2000);

}
```



d) T1, T3, T5 and T7 are in OFF conditions and T2, T4, T6 and T8 are in ON conditions simultaneously.

```
int T1 = 2;

int T2 = 3;

int T3 = 4;

int T4 = 5;

int T5 = 6;

int T6 = 7;
```



```
int T7 = 8;

int T8 = 9;

void setup() {

    pinMode(T1, OUTPUT);

    pinMode(T2, OUTPUT);

    pinMode(T3, OUTPUT);

    pinMode(T4, OUTPUT);

    pinMode(T5, OUTPUT);

    pinMode(T6, OUTPUT);

    pinMode(T7, OUTPUT);

    pinMode(T8, OUTPUT);

}

void loop() {

    digitalWrite(T1, LOW);

    digitalWrite(T2, HIGH);

    digitalWrite(T3, LOW);

    digitalWrite(T4, HIGH);

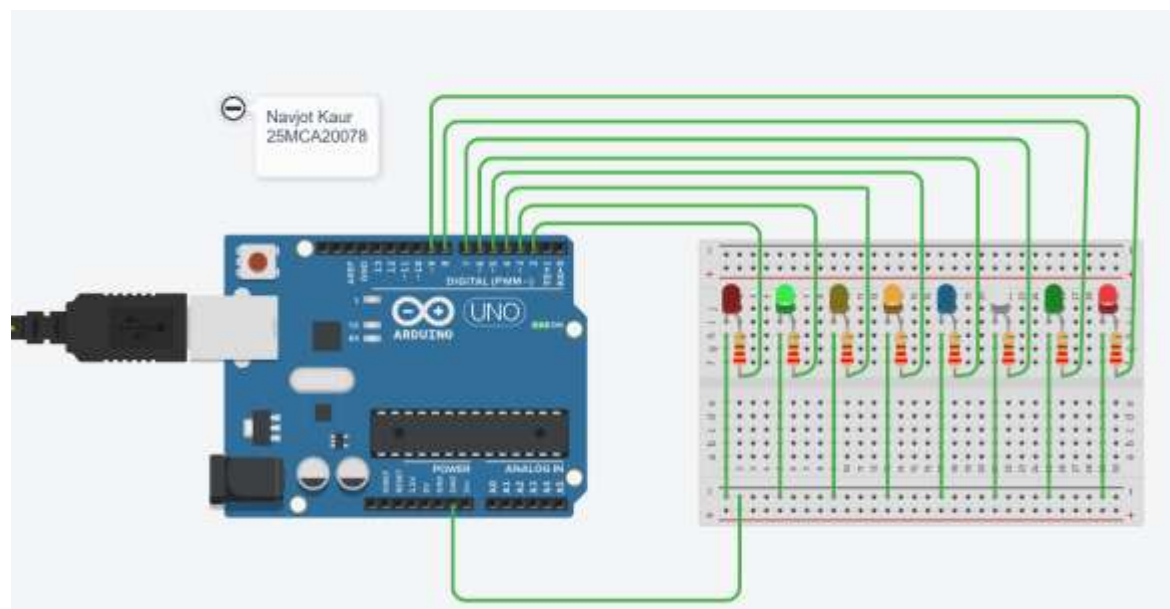
    digitalWrite(T5, LOW);

    digitalWrite(T6, HIGH);

    digitalWrite(T7, LOW);

    digitalWrite(T8, HIGH);

}
```



5. Learning outcomes (What I have learnt):

1. Learned to create a circuit.
2. Learned LED interfacing.
3. Learned Arduino programming.
4. Learned to control LEDs.
5. Learned to use `digitalWrite()` and `delay()`.